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MATLAB 3

Exercise

Consider the differential equation $y' = t/y$.

a. Use ode45 to calculate and plot an approximate solution to the initial value problem:

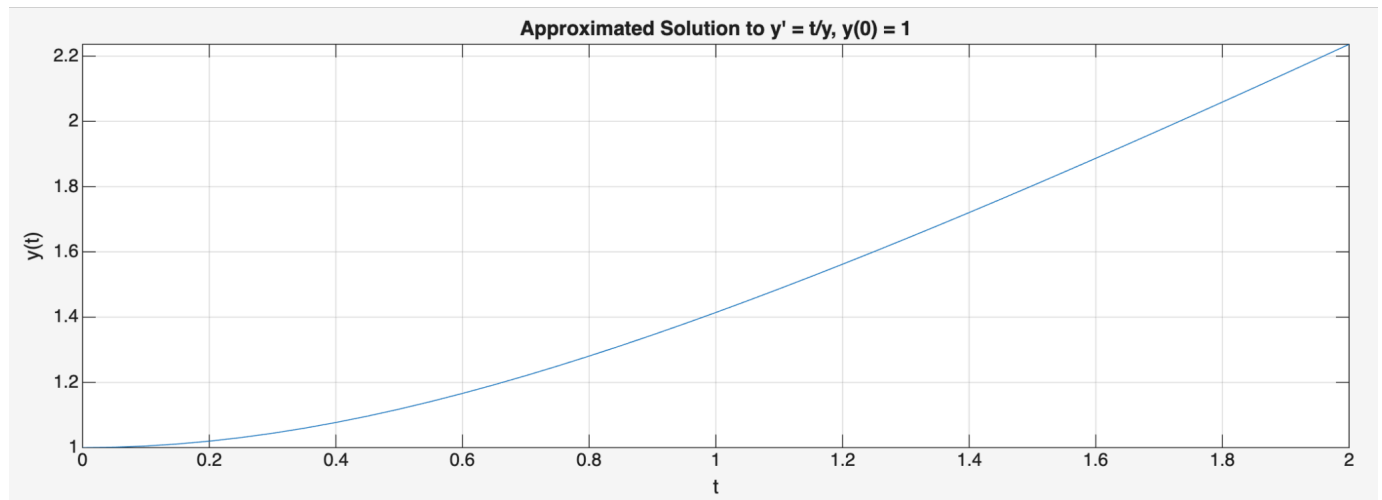
$$y' = t/y, y(0) = 1 \text{ for } 0 \leq t \leq 2$$

Command Window Screenshot

Command Window

```
>> f = @(t,y) t./y;  
>> tspan = [0 2];  
>> y0 = 1;  
>> [t, yap] = ode45(f, tspan, y0);  
>> plot(t, yap);  
>> xlabel('t');  
>> ylabel('y(t)');  
>> title('Approximated Solution to y'' = t/y, y(0) = 1');  
>> grid on;  
>> hold on  
>> axis tight
```

Plot



b. Use dsolve to find an exact solution of the IVP

Command Window Screenshot

```
Command Window
>> Sol = dsolve('Dy = t/y', 'y(0) = 1', 't')
Warning: Support for character vector or string inputs will be removed in a future release. Instead, use syms to declare variables and replace inputs such as
dsolve('Dy = -3*y') with syms y(t); dsolve(diff(y,t) == -3*y).
> In dsolve (line 126)

Sol =
(t^2 + 1)^(1/2)
```

c. Plot the exact and the approximate solution on the same graph. Can you distinguish between the two curves?

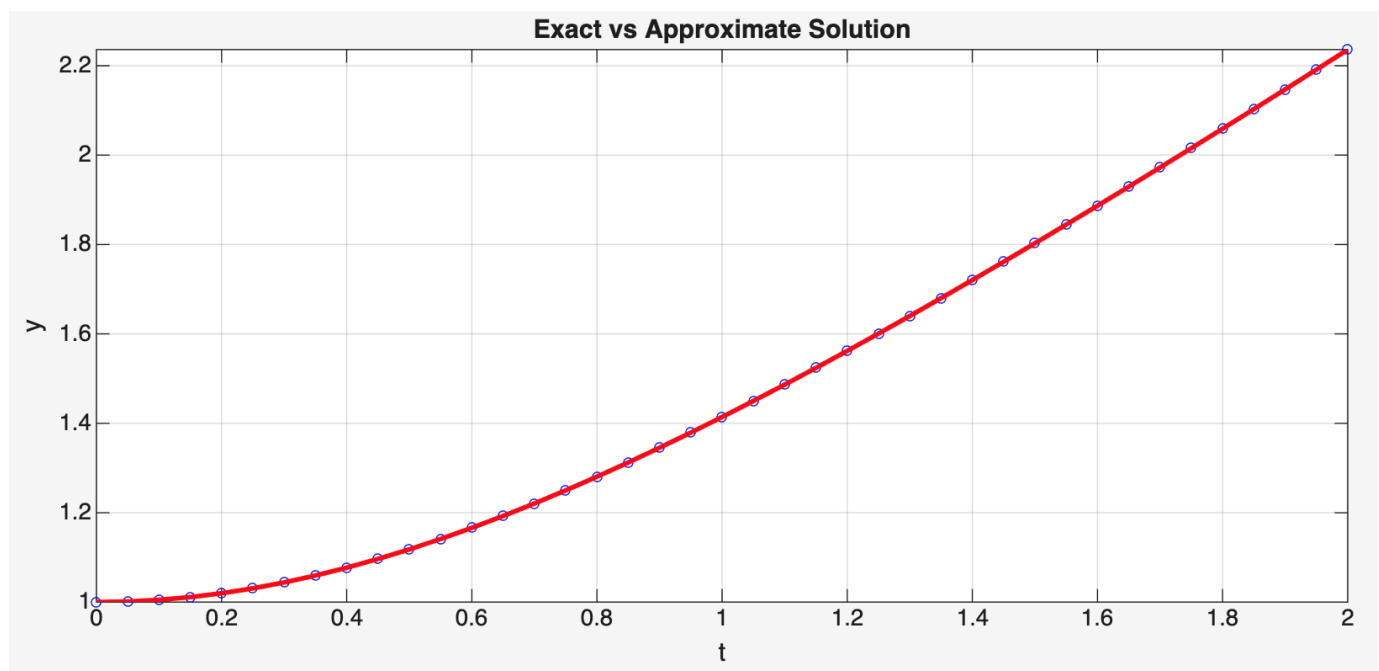
Command Window Screenshot

```

Command Window
>> f = @(t,y) t./y;
>> tspan = [0 2];
>> y0 = 1;
>> [t, yap] = ode45(f, tspan, y0);
>> syms s y(s)
>> ode = diff(y,s) == s/y;
>> cond = y(0) == 1;
>> y_exact = dsolve(ode, cond);
>> y_exact_two = sqrt(t.^2 + 1);
>> figure
>> plot(t, yap, 'bo', 'MarkerSize', 4)
>> hold on
>> plot(t, y_exact_two, 'r', 'LineWidth', 2)
>> xlabel('t')
>> ylabel('y')
>> title('Exact vs Approximate Solution')
>> grid on
>> axis tight

```

Plot



% The approximated solution essentially fully overlaps the exact solution for this ODE, so I cannot distinguish the two curves.

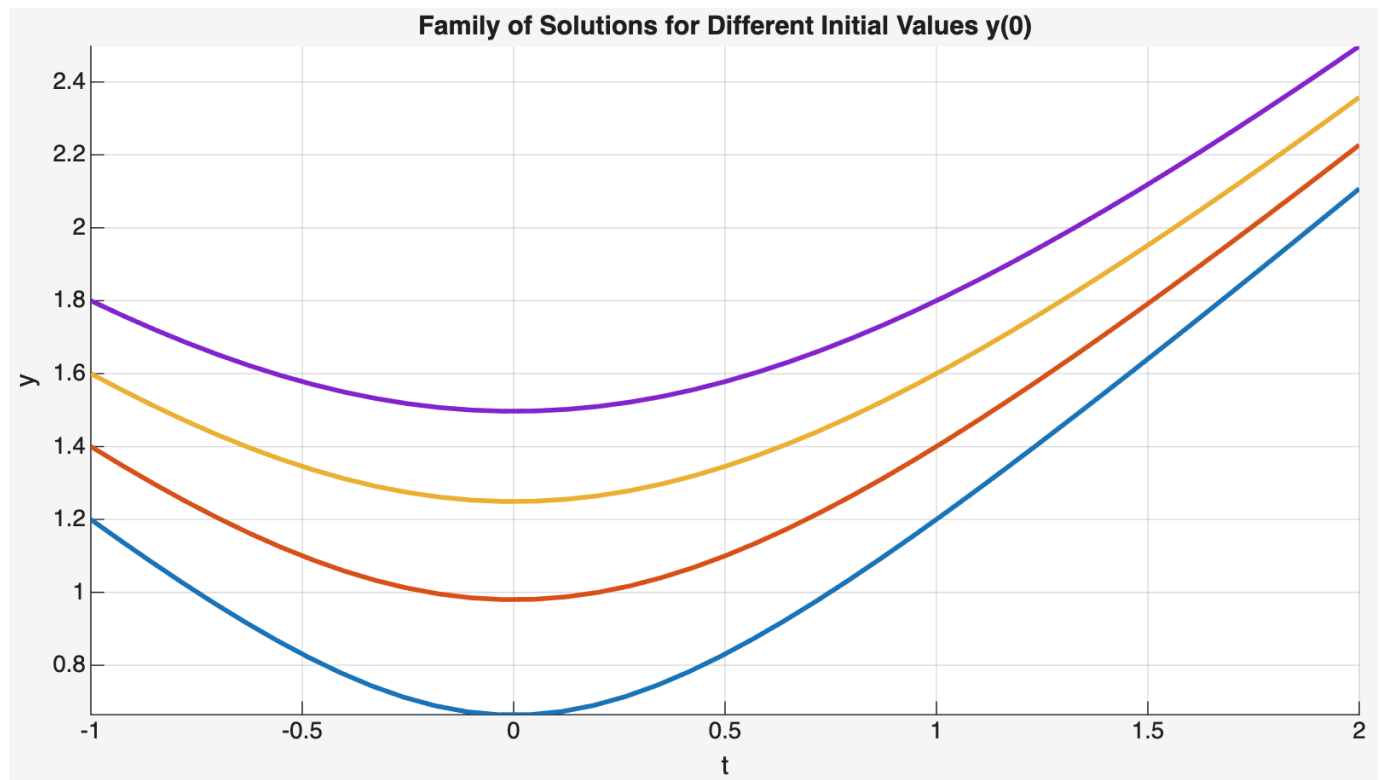
d. Plot a family of solutions with initial values $y(0) = 1.2, 1.4, 1.6, 1.8$ over the interval $[-1, 2]$

Command Window Screenshot

Command Window

```
>> f = @(t,y) t./y;  
>> tspan = [-1 2];  
>> y0_values = [1.2 1.4 1.6 1.8];  
>> [t,y] = ode45(f, tspan, y0);  
>> figure  
>> hold on  
>> for y0 = y0_values  
[t,y] = ode45(f, tspan, y0);  
plot(t, y, 'LineWidth', 2)  
end  
>> xlabel('t')  
>> ylabel('y')  
>> title('Family of Solutions for Different Initial Values y(0)')  
>> grid on  
>> axis tight
```

Plot



% Blue: $y(0) = 1.2$ Red: $y(0) = 1.4$ Yellow: $y(0) = 1.6$ Purple: $y(0) = 1.8$